

Spotted Python

Week 1 Tutorial

- ★ Course Reminders
- ★ GitHub Setup
- ★ Markdown

Welcome Spotted Pythons

- ★ Coding is an important skill for astrophysics
- ★ This stream is designed to learn coding at a slower pace
- ★ All examples and materials are the same across the 2 streams, but approached in different ways
- ★ Finn will be the tutor for Morelia



I am a small friendly snake!
If you have any questions, ask in tutorials,
on the forum or via email
(michelle.ding@unsw.edu.au)

Welcome Spotted Pythons

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Michelle (she/her)

- ★ 3rd year PhD student
- ★ Wanted to be a singer before I was 11
- ★ Decided to be an astronomer after I watched a documentary about Black Holes

Michelle (she/her)

- ★ At university I jumped around multiple research groups
 - ★ Undergrad research – pipeline to distinguish between good and bad data
 - ★ Honours research – strong lensing in group environments
 - ★ PhD research – PhD structures in edge-on galaxies
- ★ I love teaching, singing, dancing, travelling, going to markets, and seeking out good vegetarian food



I want to hear from you

- ★ What is your name
- ★ What part of astronomy are you most excited about to learn/most interested in?
- ★ Please give me feedback. I want you to govern what these tutorials are about!

Reminders

- ★ The computational assignment involves collaboration
 - ★ Each team will have a 5–10 minutes team meeting each week.
 - ★ You need to attend at least 3 meetings
 - ★ Group code must be developed on GitHub.
 - ★ You must commit lines of code and documentation for at least 3 weeks
 - ★ Commits must contain meaningful analysis
 - ★ We may ask you to explain your code to us
- ★ Responsible AI use
 - ★ AI references can be wrong
 - ★ Everything you write can be examinable
 - ★ All materials covered in the tutorials can be examinable
 - ★ We can tell AI-generated code apart from originally written code

Why use GitHub

- ★ Backup of code to reduce the chances of you losing your work
- ★ Version control
- ★ Automation tools for building and testing your work
- ★ Store and present documentation
- ★ Recognition for your work
- ★ If you are confused by this GitHub tutorial, refer to the following links (there are different ways to setup GitHub):



- ★ <https://adacs-australia.github.io/2023-03-20-Coding-Best-Practices-Workshop/GitHub/index.html>
- ★ <https://code.visualstudio.com/docs/sourcecontrol/github>

Setting up Python and GitHub

- ★ Sign up for a free GitHub account: <https://github.com/>
- ★ Install GitHub Desktop: <https://desktop.github.com/download/>
- ★ Install VS Code: <https://code.visualstudio.com/>
- ★ Install Miniconda: <https://www.anaconda.com/docs/getting-started/installation>

Making a new repository

- ★ Open GitHub Desktop
- ★ Select File → New Repository
- ★ Enter the name, location you want to save the repository, a description, and tick “Initialize this repository with a README”
- ★ Check that the file now exists locally on your laptop
- ★ Publish the repository

Publish Repository

GitHub.com GitHub Enterprise

Name
PHYS3116_2026T3

Description
Resources of PHYS3116 2026 for the Spotted Python Stream

☐ Keep this code private

Organization
None

Cancel Publish Repository

Create a New Repository ×

Name
PHYS3116_group_name

Local Path
/Users/z5309211/Work/Teaching/2026 Choose...

Description
Code and documentation for PHYS3116 computational essay

☒ Initialize this repository with a README

Git Ignore
None

× License
None

The repository will be created at /Users/z5309211/Work/Teaching/2026/PHYS3116_group_name .

Cancel Create Repository

Cloning a repository

- ★ Go to GitHub website and check your repository is published
- ★ We will now clone my repository that you will use for weekly resources
 - ★ Copy this link: https://github.com/mithebell/PHYS3116_2026T3.git
 - ★ Go back to GitHub Desktop
 - ★ File → Clone Repository → URL
 - ★ Paste in the link and clone
- ★ To push or

Clone a Repository

GitHub.com

GitHub Enterprise

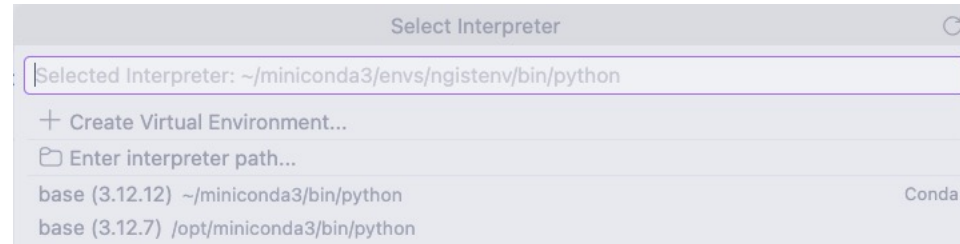
URL

Repository URL or GitHub username and repository
(hubot/cool-repo)

Local Path

Making a new Python file

- ★ Open VS Code
- ★ At the top search bar type in: >Python: Select Interpreter
 - ★ Choose the base environment from miniconda3 (it should have Conda next to it)



- ★ File → New File → Python File/Jupyter Notebook
- ★ Save Python Script or Jupyter Notebook in YOUR repository

Committing to GitHub

- ★ Open GitHub Desktop
- ★ Your new change should have shown up as a changed file
- ★ Tick the change you want to commit → Commit to main
 - ★ Here main means you are on the main branch
 - ★ Branching and forking is not required for this course

Working collaboratively on GitHub

- ★ Have one shared GitHub repository for each group
- ★ Clone GitHub Desktop onto your laptop
- ★ At the start of every coding session:
 - ★ Go to GitHub Desktop
 - ★ Repository → Pull
 - ★ Make sure your laptop is up to date with the latest version on GitHub
- ★ At the end of every coding session:
 - ★ Commit and push to GitHub so your group members can see

Markdown

- ★ Text-to-HTML conversion tool
- ★ Can be used in Jupyter Notebooks for annotation of code

Element	Markdown Syntax
Heading	# Heading 1
Bold	**Bold text** or __Bold text__
Italic	<i>*Italic text*</i> or <i>_Italic text_</i>
Highlight	==Highlighted text==
Unordered List	- First item
Link	[title](https://www.example.com)

- ★ Allows mathematical expressions in LaTeX-style syntax
- ★ Cheat sheet: <https://www.markdownguide.org/cheat-sheet/>

Practicing Markdown

- ★ Open VS Code
- ★ File → New Text File
- ★ Save the file and add the extension .md
- ★ Practice writing in Markdown and check the layout using the Markdown Editor
- ★ Commit the Markdown using instructions from previous slides